

Paul Merritt

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Work Experience

AR Development Support Specialist

NIST - New York, NY

July 2024 - Present

- Developed PLYColorParser, a Python/C# (Unity) tool that extracts and clusters colors from PLY/image files using silhouette-optimized k-means, returning dominant and WCAG-contrast-safe colors for UI design.
- Upgraded ARTV to Quest 3 with new GUIs and point cloud models.
- Built XRDM, a WebXR A-Frame experience embedded in Qualtrics to collect decision-making metrics in survey studies.
- Presented at FedCASIC 2025 on using WebXR to enhance remote research studies, showcasing our XRDM platform for immersive data collection within survey environments.

Technical Artist

Meta - New York, NY

October 2022 - April 2024

- Co-developed a variety of projects for Meta, contracted through Insight Global. Project details, purpose, and technologies are bound by NDA.
- Collaborated on AR filters and Snap lenses.
- Implemented features at request of project managers, analyzed and optimized software performance, and pitched solutions to UX concerns.
- Regularly participated in on-call rotation 3 days at a time.
- Assisted in triaging and debugging wearable XR device features, contributing to internal testing pipelines and Android-side UX improvements.

Freelance AR Developer

Bolder Games - Remote

August, 2022 - September 2022

- Worked on a small team of developers operating in a SCRUM workflow on an Augmented Reality project. This project was for The Walt Disney Company, contracted through Bolder Games. Project details, purpose and technologies are bound by NDA.

AR Developer

NIST - Boulder, CO

September, 2018 - October, 2022

- Developed ARTV (Augmented Reality Tracking Visualization) which displays real-time tracking on 3D point clouds.
- Designed and co-designed several WebXR prototypes that stream IoT data via MQTT to an A-Frame scene display.
- Co-facilitated the CHARIoT Challenge, bridging IoT and AR industries towards improving a First Responder's performance.
- Used C# and Unity to build Augmented Reality (AR) programs for The National Institute of Standards and Technology's (NIST) Public Safety Communications Research Division (PSCR).
- Created and demoing POC-level training simulations for the Hololens and Magic Leap.
- Developed PSCR ML Logging, a multithreaded Magic Leap logging system that streamed real-time gesture, input, gaze, and camera data from the device to support behavioral and spatial analytics in AR simulations.
- Produced scenarios for the Vuzix Blade and iOS/Android to showcase the potential of AR in a variety of form factors.
- Constructed 3D models and their animations for the iOS/Android app.
- Worked alongside local and distant Fire Fighters and Bomb Disposal experts to isolate key needs in their fields.
- Collaborated with cross-functional engineers to identify, reproduce, and resolve issues in device firmware and companion Android applications.
- Developed internal Android utilities and UI features to streamline wearable testing.
- Managed feature implementation and validation timelines independently, coordinating across design and engineering teams to maintain lab readiness.
- Applied AOSP experience to customize builds for prototype hardware, ensuring compatibility and performance across test environments.

Summary

Paul Merritt graduated NYU in 2018 and began working at NIST in the PSCR division. He has completed several projects in Augmented Reality as part of the UI/UX portfolio. They're simulations intended for firefighters, bomb squads, and police officers: for the Microsoft Hololens, the Magic Leap One, the Vuzix Blade, and WebXR. He co-led and judged the CHARIoT Challenge, bringing together industry and academia to create innovations in HUDs. After, he completed a AR freelance contract with Bolder Games for The Walt Disney Company. He went on to work as a Technical Artist at Meta for 2 years before returning to NIST PSCR to continue supporting first responders through advancements in XR technology. He is experienced in Android development and embedded device testing, including debugging wearable hardware, implementing AOSP-level features, and supporting lab environments through rapid issue triage and root-cause analysis.

Skills

Languages & Tech Stacks

TypeScript	Java
React Native	C++
A-Frame	C
AR.js	JavaScript
p5.js	Node.js
ML5.js	HTML5
C#	CSS
	Python

Environments

Microsoft Azure AI	MS Office
Meta Spark Studio	Adobe Premiere
Unity	Adobe Photoshop
Maya	Audacity
Git	OBS
MQTT	

Platforms

Meta Quest 3	iOS
Microsoft HoloLens	Android
MagicLeap One	Web
Vuzix Blade	

Certifications

Azure AI Engineer Associate

May, 2025

Education

New York University, New York, NY

May, 2018

Bachelor's Degree: Computer Science

Minor: Game Engineering